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Fabrication of surface microstructures by mask electrolyte jet machining

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ABSTRACT

This paper presents a novel electrochemical processing technique, mask electrolyte jet machining (MEJM), for the fabrication of surface microstructures. MEJM combines jet electrochemical micromachining (Jet-ECM) and through-mask electrochemical micromachining (TMEMM), combining the advantages of TMEMM, which is a high-throughput process, and of Jet-ECM, with its adjustable flow field. The effects of a mobile nozzle on electrolyte flow are investigated, and a new modeling approach for large translational movements is proposed. An analysis of the accuracy and reliability of the proposed method is presented. Microprotrusions and microdimples are produced to high precision and with excellent consistency of dimensional variation (maximum standard deviation 2.171 μm). The results suggest the possibility of an affordable technique for batch fabrication of surface microstructures with high efficiency and precision.

1. Introduction

In the last few decades, there has been increasing interest in the application of surface structures, particularly on the micro- and nanoscale, owing, for example, to their tribological and wetting properties [1, 2]. A variety of approaches to the fabrication of micro/nanoscale surface structures have been investigated. Electrochemical micromachining (ECM) is widely employed [3–7]. This is a noncontact and nonthermal process and thus does not lead to tool wear or to mechanical and thermal residual stresses that might initiate changes in material properties [8,9]. The material removal rate can be adjusted by controlling the applied current. A wide range of metallic materials, regardless of their hardness and thermal resistance, can be easily machined by ECM.

For the fabrication of surface structures, a number of techniques based on ECM are available, including jet electrochemical machining and through-mask electrochemical micromachining. Jet electrochemical machining (Jet-ECM) employs a jet of electrolyte: a stream of electrolyte flows through a nozzle and impinges directly on the workpiece, so that the electrochemical reaction is localized to a small region. Sen and Shan [10] investigated the effects of process parameters such as applied voltage, nozzle diameter, and electrolyte pressure on the quality of small holes produced by Jet-ECM. This technique has demonstrated its capacity to obtain various shapes down to the micrometer range [11], even its capability of fabricating microstructures on metal carbides [12]. However, Jet-ECM suffers from a relatively slow speed because of the need for sequential processing. Costa and Hutchings [13] employed textured tools consisting of arrays of holes to fabricate microdimples by Jet-ECM in a more efficient parallel machining process. However, the shape, size, and precision of microfeatures in Jet-ECM largely depends on the size of the nozzle: to generate smaller features, the inner diameter of nozzle must be reduced. Not only does this increase manufacturing costs, but it is extremely difficult to maintain a sustainable and stable electrolyte flow on a microscale. In addition, to generate different microsurface structures, such as triangles and squares, it is necessary to use nozzles of the corresponding shape, which poses further problems for nozzle manufacture. This is a natural limitation on the implementation of Jet-ECM for microfabrication.

Through-mask electrochemical micromachining (TMEMM), another common method, employs photolithography to produce micropatterns on photoresist coated on the workpiece surface, so that a large number of desired areas dissolve in parallel. This technique is precise and relatively fast, capable of generating well-defined surface textures with controlled size, location, and density. Using this method, Wang et al. [14] fabricated three-dimensional cylindrical microstructures with feature sizes as small as 40μm. Landolt et al. [15] carried out TMEMM of titanium using a laser-patterned oxide film. Qu et al. [16] developed a modified TMEMM technique for fabricating microdimple arrays with a re-useable polydimethylsiloxane mask. Ming et al. [17] used a patterned inert metal plate to fabricate metal through-hole arrays with double tapered openings. Qian et al. [18] were able to achieve a significant

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Tackling data scarcity in machine learning-based CFRP drilling performance prediction through a broad learning system with virtual sample generation (BLS-VSG)

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ABSTRACT

Machine learning (ML)-based data-driven method has emerged as a powerful tool for predicting the manufacturing performance of carbon fibre reinforced plastic (CFRP), particularly in CFRP machining, where physics-based models are computationally expensive. However, the effectiveness of ML models are often constrained by limited datasets, due to the high cost and time required for experimental data acquisition. To address this, this paper presents the first study to apply virtual sample generation (VSG) techniques to enlarge the training dataset and mitigate data scarcity in the prediction of CFRP drilling performance. A novel hybrid ML framework integrating Broad Learning System (BLS) and VSG (BLS-VSG) is proposed to combine the capability of BLS in small dataset prediction with the enlarged dataset generated by VSG. The model has been employed to predict the drilling thrust force and delamination damage under various drilling conditions (spindle speed, feed rate, point angle). Three different VSG methods (SMOTE, MD-MTD and CVT) and the number of virtual samples were evaluated in detail. Results show that VSG can effectively enlarge the training dataset and improve the prediction performance of the ML model. Specifically, VSG reduced the mean square error (MSE) and mean absolute percentage error (MAPE) for thrust force prediction by 39.0 % and 12.9 %, respectively, compared to the benchmark without VSG. For delamination factor F_{da} prediction, MSE and MAPE were reduced by 22.6 % and 16.5 %, respectively. The proposed BLS-VSG model outperforms other conventional ML models (BPNN, ELM, SVR and RT) for both scenarios (with/without VSG), providing a robust and data-efficient solution for CFRP drilling performance prediction.

Nomenclature

F_{da}	Delamination factor
D_{max}	Maximum diameter of delaminated area
D_{nom}	Nominal diameter of the hole
A_{max}	Area related to the maximum diameter of the delaminated zone (D_{max})
A_{nom}	Nominal hole area
A_{dam}	Delaminated area
$\mathbf{X}$	Input matrix of original dataset
$\mathbf{Y}$	Output matrix of original dataset
N	The number of samples in the original dataset
D	The number of attributes

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$\boldsymbol{x_r}$	Selected root sample for SMOTE
$\boldsymbol{x_{rk}}$	k nearest neighbours of root sample $\boldsymbol{x_r}$
d	Euclidean distance
$\boldsymbol{x_{ra}}$	Randomly selected auxiliary sample for SMOTE
$\boldsymbol{x_{vir}}$	Generated virtual sample
CL_j	Data centre point (median) for attribute j
$x_{[-]j}$	Ordered dataset after sorting for attribute j
Skew_{Lj}	Left skewness
Skew_{Uj}	Right skewness
$\widehat{S}_j^2$	Variance

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Research paper

Data-driven models with physical interpretability for real-time cavity profile prediction in electrochemical machining processes

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ABSTRACT

The Electrochemical Machining (ECM) process can be effectively controlled by adjusting the process parameters, including electrolyte composition, current density, etc. However, this control strategy presents challenges associated with modeling the complex relationships between processing parameters and the resulting process outcomes. Data-driven approaches hold promise for real-time cavity profile prediction, while their black-box nature limits the interpretability. To address these issues, this study initiated by developing several Machine Learning (ML) models to predict cavity profiles using process parameters, in-process data, or a combination of both. Subsequently, these models were inspected by explainable artificial intelligence (XAI) methods. Using process parameters for model inspection provides preliminary guidance for achieving desired outcomes, while inspections with in-process data shed light on process dynamics and enable diagnostics by detecting anomalies through atypical model focus. Linear regression (LR), with high interpretability, offers moderate predictive accuracy, whereas neural networks (NN) and convolutional neural networks (CNN) perform better yet require XAI methods to interpret their decision-making mechanisms. Model inspection was conducted on global and local levels. Global inspection via SHapley Additive exPlanations (SHAP) for LR and NN models identified factors affecting cavity size, aligning predictions with ECM knowledge. Local inspection through Gradient-weighted Class Activation Mapping (Grad-CAM) examined CNN predictions, revealing temporal process dynamics and model focus during specific processing stages. The impact of unexpected events identified by the ML models was validated using a physics-based model. A web application was developed to integrate these insights, allowing for real-time cavity prediction and visualization based on process parameters and in-process data.

1. Introduction

The recent integration of machine learning (ML) techniques has improved operational efficiency and precision in various manufacturing processes. However, a critical aspect of scientific investigations remains important: the development and expansion of knowledge beyond empirically derived rules. While ML offers practical solutions, the underlying basis for its decision-making can sometimes be a "black-box", opaque to engineers and scientists. This opacity presents substantial challenges, especially in critical manufacturing contexts where comprehension and confidence in AI-derived outputs are essential. Concomitant advancements in sensor technology and data acquisition capabilities have led to the generation of vast amounts of in-process data during manufacturing operations (Dogan and Birant, 2021). These in-process data streams offer invaluable insights into various facets of production, including the production environment, equipment performance, and product quality. By effectively leveraging these data streams, it becomes possible to identify complex, non-linear relationships across diverse data types and sources. This paves the way for enhanced decision-making and process optimization within manufacturing.

Electrochemical machining (ECM) is a widely adopted non-conventional manufacturing process employed in diverse industries for the fabrication of components used in aerospace engines (Xu and Wang, 2021), medical equipment (Hung and Yang, 2021), automotive parts (Demirtas et al., 2018), and electronic devices (Koyano et al.,

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Intelligent discharge state detection in micro-EDM process with cost-effective radio frequency (RF) radiation: Integrating machine learning and interpretable AI

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ABSTRACT

Machining state monitoring is an enabling technology that can ensure the smoothness of production efficiency and quality by indirectly understanding various discharge patterns and anomalous events. The predominant method for identifying discharge pulses in micro electrical discharge machining (micro-EDM) relies on electrical sensing near the machining zone, which may interfere with the process. To provide a complementary approach further from the machining zone and independent from machine wiring, this paper proposes a discharge state detection scheme for micro-EDM using cost-effective and non-invasive radio frequency (RF) signals during sparking. A threshold-based method is first employed for preliminary pulse classification, followed by manual correction to ensure data reliability. Based on RF signal characteristics, domain knowledge is incorporated to extract four feature categories as inputs for machine learning (ML) models. Tree-based ensemble learning methods are applied to predict discharge pulses, with XGBoost attaining the highest accuracy. SHapley Additive exPlanations analysis reveals that distribution features of RF signals contribute most to pulse recognition, while statistical features have a smaller impact. Furthermore, ML-based classifications are compared with deep learning models. The designed hybrid architecture combining temporal convolutional networks (TCN) and gated recurrent units, can improve the identification accuracy across all pulse types, with a macro F1-score larger than 94 %. Gradient-weighted class activation mapping shows that TCN modules emphasize the significance of information on dielectric breakdown and discharge maintenance stages. Despite potential data loss from reduced sampling rates, key signal characteristics remain intact, with prediction accuracy degradation below 1 %. This study demonstrates the feasibility of RF-based discharge state detection, offering an alternative to electrical sensing and providing insights for improving servo control and machining efficiency in future EDM processes.

1. Introduction

Although micro electrical discharge machining (micro-EDM) accounts for a small market share in the entire manufacturing industry, its thermoelectrically coupled material removal mechanism gives a natural benefit in processing difficult-to-cut materials, making it an indispensable manufacturing solution. The miniaturization of components has become a progressive trend in the sectors of electronics, medicine, aerospace, etc., which also promotes the prosperity of micro-EDM (Leo Kumar et al., 2014). However, this machining technique has strong nonlinear and dynamic characteristics mainly due to the involvement of transient multi-physical phenomena, which makes the discharge process exhibit randomness and uncertainty, thereby hindering the actual optimization of efficiency and accuracy. Consequently, it is essential to deploy corresponding sensors and measurement systems in machine tools to monitor the machining process. Unlike traditional force-dominated cutting monitoring strategies, EDM monitoring primarily refers to identifying various discharge states within the gap and implements preventive measures and adjustments through discovering anomalies (Zhao et al., 2025). In recent years, thanks to the development of big data, artificial intelligence (AI), and advanced sensing technology, production efficiency and quality can be significantly improved by data-driven decision-making and real-time monitoring systems, which also provide possibilities for stable and effective EDM processing.

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